
Fundamentals of Information Systems Security

Fourth Edition

Project 1

Comprehensive Risk Assessment

Student Name: _____

Student ID: _____

Instructor: Guuleed Maxamuud Aw Abdi

Date: _____

Weight: 20%

Submission Format: PDF Report via Course Platform

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1 Project Overview

About This Project

In this project, you will conduct a comprehensive risk assessment for a fictional organization. You will apply the principles and methodologies covered in **Chapters 3 and 4** to identify assets, threats, vulnerabilities, and risks, and then propose appropriate risk treatment strategies.

The goal is to demonstrate your understanding of the risk management lifecycle and your ability to produce a professional risk assessment report suitable for management review.

2 Learning Objectives

Upon completion of this project, you will be able to:

1. Identify and classify information assets within an organization
2. Identify relevant threats and vulnerabilities across all seven IT domains
3. Analyze and prioritize risks using both quantitative and qualitative methods
4. Propose appropriate risk treatment options (accept, mitigate, transfer, avoid)
5. Prepare a professional risk assessment report suitable for management review

3 Scenario

Organization Profile

You have been hired as a security consultant for **TechSolutions Inc.**, a mid-sized software development company with 250 employees. The organization has three main locations: headquarters (HQ), a development center, and a small branch office.

Key Business Operations:

- Software development and distribution
- Customer support and technical assistance
- Cloud-based services for clients
- Financial and HR operations

3.1 IT Infrastructure

The IT infrastructure consists of the following seven domains:

highlightblue Do-main	Components	Security Concerns
User Domain	250 employees with laptops and mobile devices	Social engineering, weak passwords
Workstation Do-main	Company-issued Windows laptops, macOS	Malware, unauthorized software
LAN Domain	Switches, routers, internal servers	Eavesdropping, unauthorized access

LAN-to-WAN Domain	Firewalls, VPN concentrator, DMZ	Perimeter breaches, intrusion
WAN Domain	Internet via two ISPs	Data interception, DDoS
Remote Access Domain	VPN for remote workers	Unsecured connections, credential theft
System/Application Domain	Web servers, database servers, file servers, Microsoft 365	Vulnerabilities, data breaches

4 Project Tasks

4.1 Task 1: Asset Inventory and Classification

Task 1 Instructions

1. Identify all information assets relevant to TechSolutions Inc.
2. Classify each asset according to sensitivity (Public, Internal, Confidential, Restricted)
3. Assign an asset value (monetary or qualitative) and criticality level

4.2 Task 2: Threat and Vulnerability Identification

Task 2 Instructions

1. Identify at least 10 distinct threats that could affect the organization
2. For each threat, identify the vulnerabilities that could be exploited
3. Map threats to the affected assets and domains

4.3 Task 3: Risk Analysis

Task 3 Instructions

1. Perform both quantitative and qualitative risk analysis
2. Calculate SLE, ALE, and ARO for at least three significant risks
3. Use a risk matrix to prioritize risks
4. Create a risk register with all identified risks

4.4 Task 4: Risk Treatment Recommendations

Task 4 Instructions

1. For each prioritized risk, propose appropriate treatment options
2. Justify your recommendations with cost-benefit analysis where possible
3. Describe the controls to be implemented

5 Deliverables

You must submit a professional report containing the following sections:

1. **Executive Summary** (1 page)
2. **Introduction** (context and scope)
3. **Asset Inventory and Classification** (table)
4. **Threat and Vulnerability Analysis** (narrative and tables)
5. **Risk Analysis Results** (risk register, risk matrix, quantitative calculations)
6. **Risk Treatment Plan** (recommended controls, timelines, responsibilities)
7. **Conclusion and Recommendations**
8. **Appendices** (detailed calculations, additional data)

6 Evaluation Rubric

Criteria	Weight	Description
Asset Identification and Classification	15%	Complete and accurate identification of assets with proper classification
Threat and Vulnerability Identification	20%	Comprehensive identification of relevant threats and vulnerabilities
Risk Analysis (Quantitative and Qualitative)	25%	Proper calculations, risk register, and risk matrix
Risk Treatment Recommendations	20%	Well-justified, feasible recommendations
Report Quality and Professionalism	10%	Clarity, structure, grammar, and formatting
Critical Thinking and Justification	10%	Depth of analysis, logical reasoning, and creativity

7 Submission Guidelines

Submission Requirements

- Submit a single PDF file via the course platform
- File name: **Project1_RiskAssessment_YourName.pdf**
- Use 12pt font, 1-inch margins, and consistent formatting
- Include a title page with your name, course, and date
- Cite any references used (e.g., NIST SP 800-30, ISO 27005, textbook)

8 Due Date and Late Policy

Important Deadlines

- **Due Date:** _____ (Week 8)
- **Late Policy:** Submissions will be penalized 10% per day
- **Extensions:** Must be requested at least 48 hours in advance

9 Resources

- NIST SP 800-30 (provided in frameworks folder)
- Textbook Chapters 3 and 4
- Risk assessment template (provided in policy-templates)
- Online tutorials on risk matrices and ALE/SLE calculations